

8259 Programmable Interrupt Controller (PIC) MPCA Module 4

> **Definition:** The **Intel 8259A (PIC)** is an 8-channel programmable interrupt controller designed to manage, prioritize, and translate multiple hardware interrupt requests (**IR_0 - IR_7**) into 8086 vector interrupt instructions.

1. Detailed Technical Explanation

Internal Architecture & Block Diagram of 8259

8259 I

2. 8259 Interrupt Processing Sequence

1. Peripheral asserts an interrupt line (**IR_0 - IR_7**).
2. **IR**

3. Priority Modes in 8259

1. **Fully Nested Mode (Default):** **IR_0** has the highest priority and **IR_7** has the lowest priority. Lower priority interrupts are

inhibited while a higher priority ISR is running.

2. **A

4. Must-Write Points for Exams

- **IRR** stores requested interrupts, **ISR** stores currently executing interrupt, **IMR** masks unwanted interrupts.

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5. Quick Recall Flow

IR0-IR
Comm